ENDG 319

Probability, Statistics and Machine Learning

Lecture 11: Normal and Lognormal Distributions

Standard units • Normal-curve areas • Linear functions
Assessing normality • The lognormal transformation

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Learning outcomes for Lecture 11

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FROM A DISTRIBUTION MODEL TO A PROBABILITY OR A MEASUREMENT

1	**Describe**	Read $N(\mu, \sigma^2)$ and interpret its centre and spread.
2	**Standardize**	Move between original measurements and z-scores.
3	**Calculate**	Use left-tail areas, complements, and percentile lookups.
4	**Transform**	Find distributions of linear functions and sample means.
5	**Assess**	Recognize the normal–lognormal relationship and its limits.

> Name the variable, state the model, and sketch the requested area before calculating.

The normal distribution: read the parameters correctly

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THE SECOND PARAMETER IN N(μ, σ²) IS A VARIANCE

$$X \sim N(\mu, \sigma^2), \qquad \sigma > 0$$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \quad -\infty < x < \infty$$

μ — mean

Sets the centre of the curve.

σ — standard deviation

Spread, in the original units.

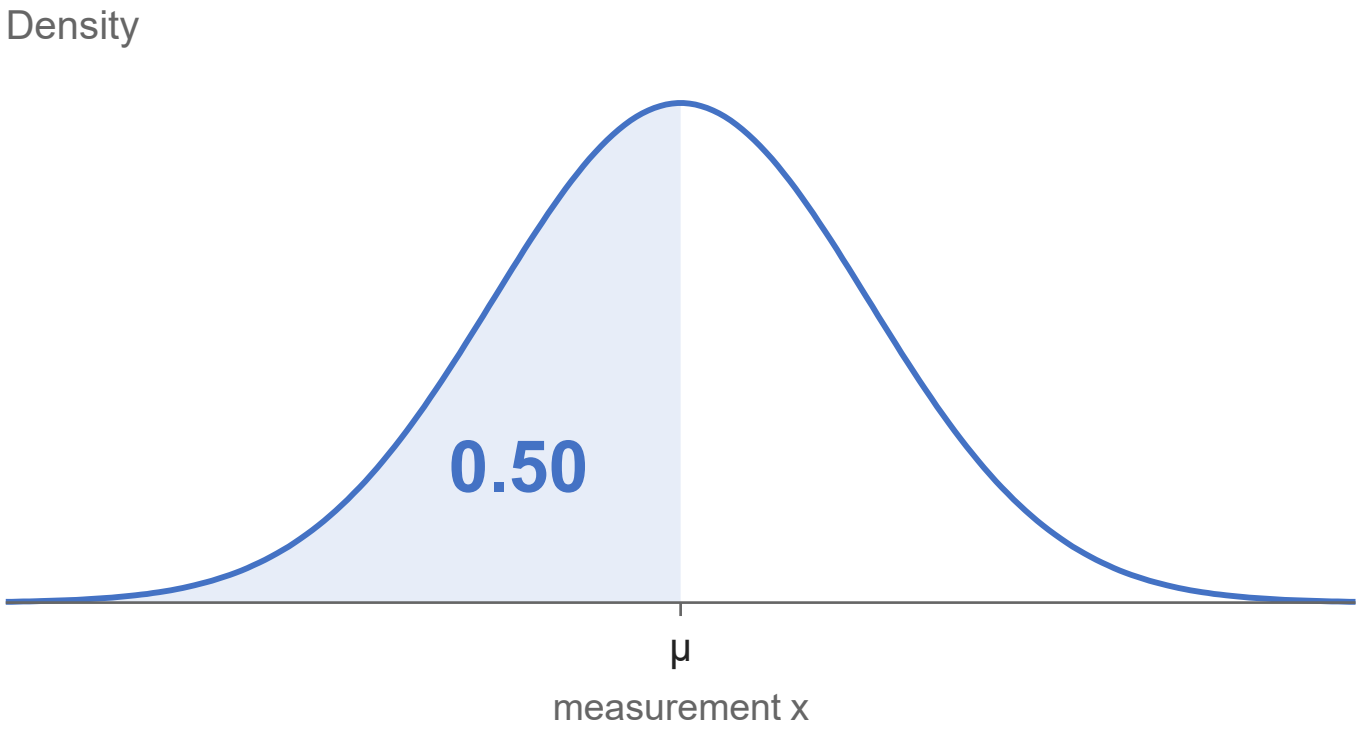
σ² — variance

Spread in squared units.

Given a variance, take its square root before using the standardization formula.

Symmetry locates the centre and splits the probability

μ IS BOTH THE MEAN AND THE MEDIAN



$$P(X < \mu) = \frac{1}{2}$$

$$P(X > \mu) = \frac{1}{2}$$

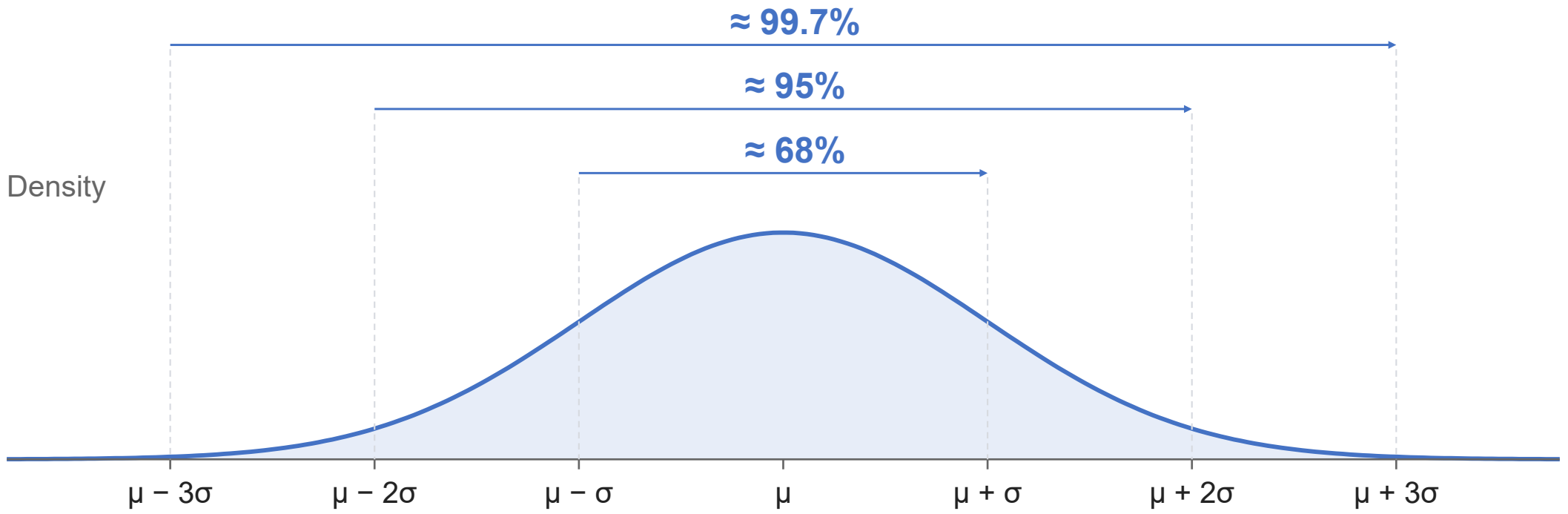
Symmetric about μ.

$$P(X = x) = 0$$

A single exact value has zero probability; probabilities are areas over intervals.

The 68–95–99.7 rule is a normal-model approximation

THE INTERVALS ARE CENTRED AT μ AND MEASURED IN STANDARD DEVIATIONS



These are population proportions under a normal model—not guarantees for every sample.

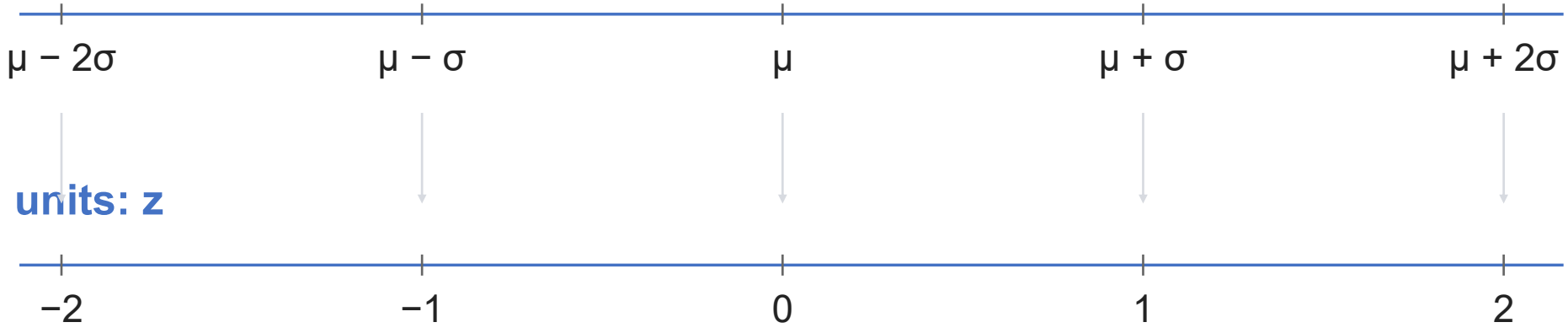
Standard units: subtract the mean, divide by the SD

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STANDARDIZATION CHANGES THE SCALE—THE NORMAL ASSUMPTION COMES FIRST

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Original units: x



A z-score is a signed number of standard deviations from μ; it has no measurement units.

Example 7: aluminum-sheet thicknesses

PAUSE AND IDENTIFY WHAT IS GIVEN AND WHAT IS REQUESTED

Thickness X is measured in thousandths of an inch.
The normal population has mean 10 and standard deviation 1.3.

$$X \sim N(10, 1.3^2)$$

(a) Original units → standard units

Thickness: 10.8 (in the stated units).
Find its z-score.

(b) Standard units → original units

A sheet has z = −1.7.
Find its thickness in the original units.

Predict the sign or location of the answer before doing the arithmetic.

Example 7: convert in both directions

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WORKED SOLUTIONS DERIVED FROM THE SUPPLIED INPUTS

(a) A measured thickness gives a z-score

$$z = \frac{10.8 - 10}{1.3} = \frac{0.8}{1.3} \approx 0.6154$$

The sheet is about 0.62 standard deviations above the mean.

(b) A z-score gives a measured thickness

$$x = 10 + 1.3(-1.7) = 7.79$$

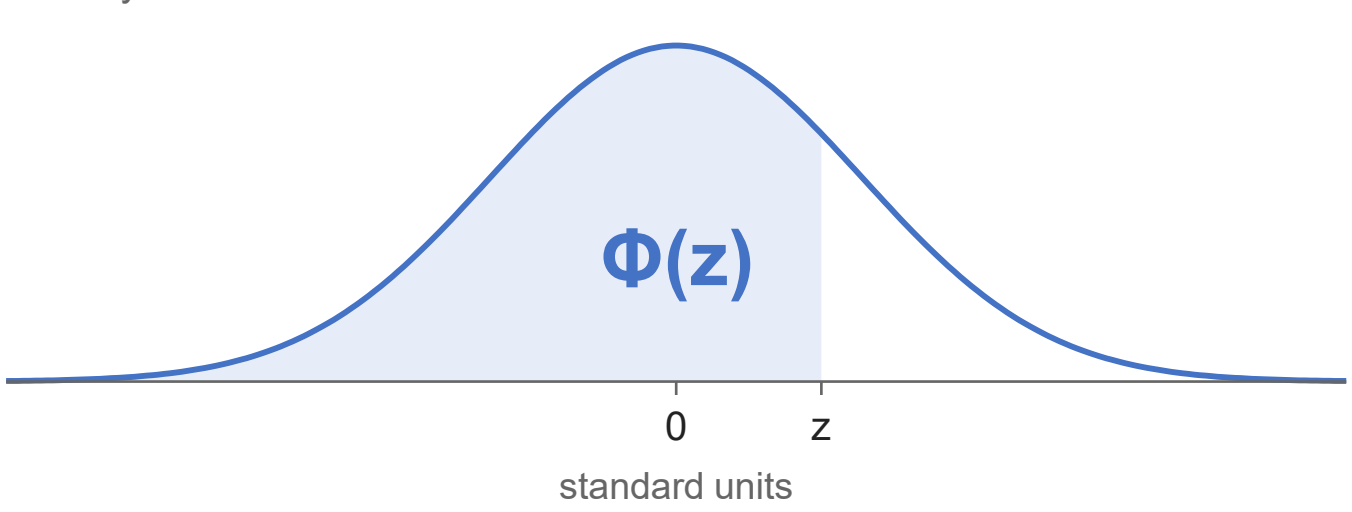
Thickness = 7.79 thousandths of an inch, below the mean.

> Do not add units to z; restore the original measurement units when reporting x.

Normal probabilities are cumulative areas

THE STANDARD NORMAL TABLE USED HERE GIVES LEFT-TAIL AREAS

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \frac{e^{-t^2/2}}{\sqrt{2\pi}}\, dt$$



Read Φ(z) as:
“probability at or below z.”

Use a table or numerical calculation for the area.
PDF height is not probability.

The symbol Φ is a cumulative probability; the normal PDF gives a density height.

Read a left-tail z-table: row plus column

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ILLUSTRATION USING THE SOURCE VALUE z = 0.47

z = 0.47 = 0.4 + 0.07

z	0.05	0.06	0.07	0.08	0.09
0.3	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6736	0.6772	**0.6808**	0.6844	0.6879
0.5	0.7088	0.7123	0.7157	0.7190	0.7224

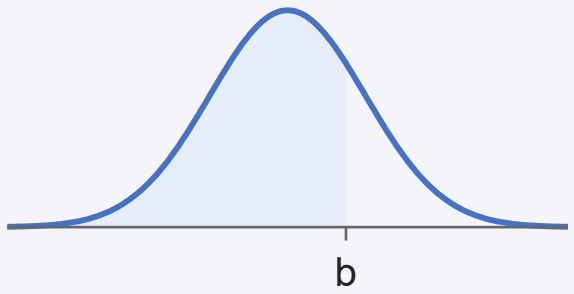
Row 0.4 and column 0.07 intersect at 0.6808.

Table A.2 is referenced in the original notes; this small classroom excerpt is computed.

Three area requests use one cumulative table

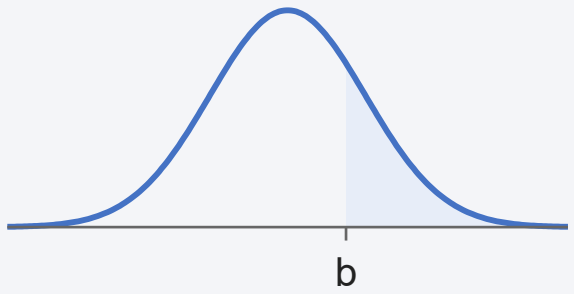
SKETCH FIRST, THEN CHOOSE DIRECT LOOKUP, COMPLEMENT, OR DIFFERENCE

Left of b



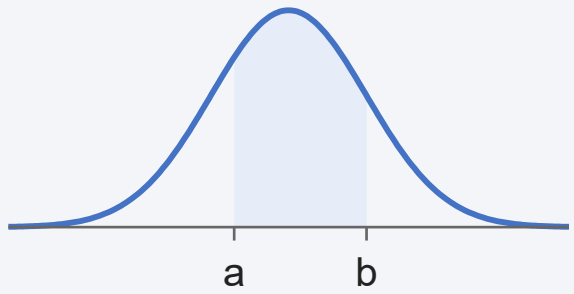
$$P(Z \leq b) = \Phi(b)$$

Right of b



$$P(Z > b) = 1 - \Phi(b)$$

Between a and b



$$P(a < Z < b) = \Phi(b) - \Phi(a)$$

For the continuous normal model, < and ≤ give the same interval probabilities.

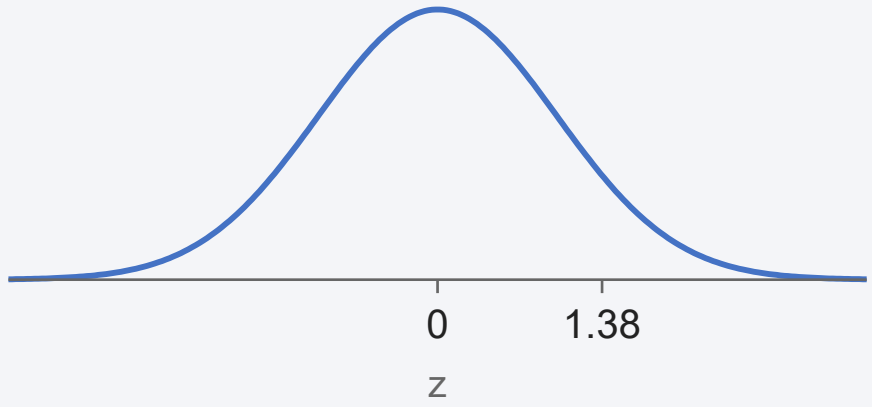
Example 8: choose the correct tail

IDENTIFY THE SHADED SIDE BEFORE LOOKING UP A NUMBER

(a) Area to the left of z = 0.47



(b) Area to the right of z = 1.38



Without calculating: which requested area should be larger than 0.5?

Example 8(a): area to the left of z = 0.47

DIRECT LEFT-TAIL LOOKUP

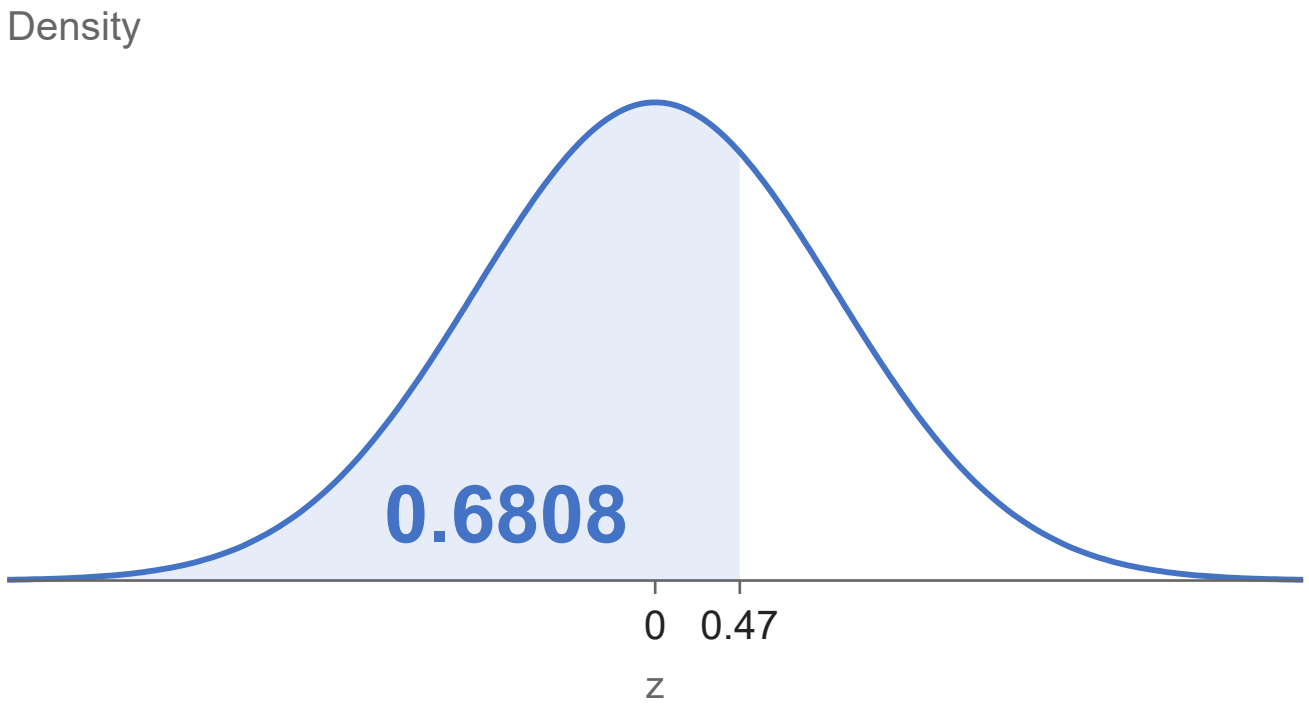


Table entry = requested area

$$P(Z < 0.47)$$

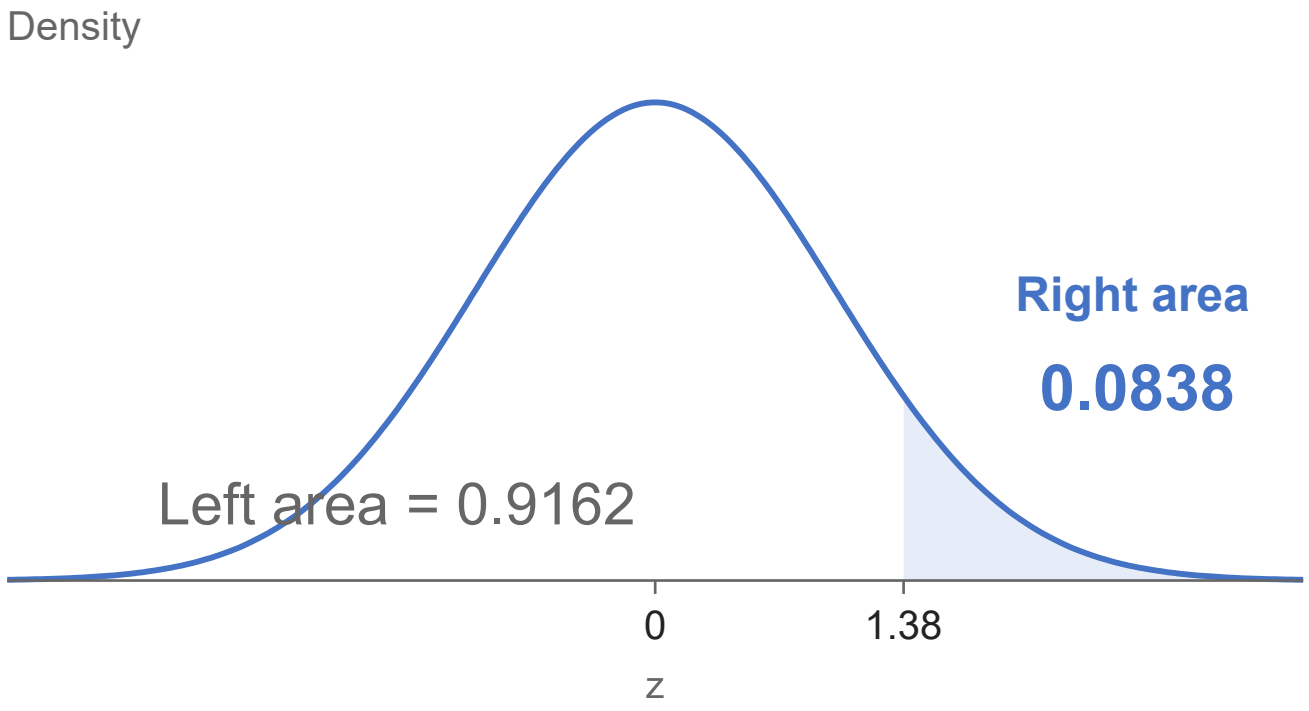
$$= \Phi(0.47) \approx 0.6808$$

About 68.08% lies below z = 0.47.

Sanity check: 0.47 is above the mean, so its left-tail area must exceed 0.5.

Example 8(b): area to the right of z = 1.38

COMPLEMENT THE LEFT-TAIL TABLE ENTRY



Total area = 1

$$P(Z > 1.38) = 1 - \Phi(1.38)$$

$$\approx 1 - 0.9162 = 0.0838$$

About 8.38% lies above 1.38.

The lookup 0.9162 is not the answer to a right-tail question; subtract it from 1.

Example 9: an interval and a reverse lookup

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TWO DIFFERENT TASKS: FIND AN AREA, OR FIND A LOCATION

(a) Find the area between z = 0.71 and z = 1.28.

Write the difference of two left-tail areas in the correct order.

(b) What z-score corresponds to the 75th percentile?

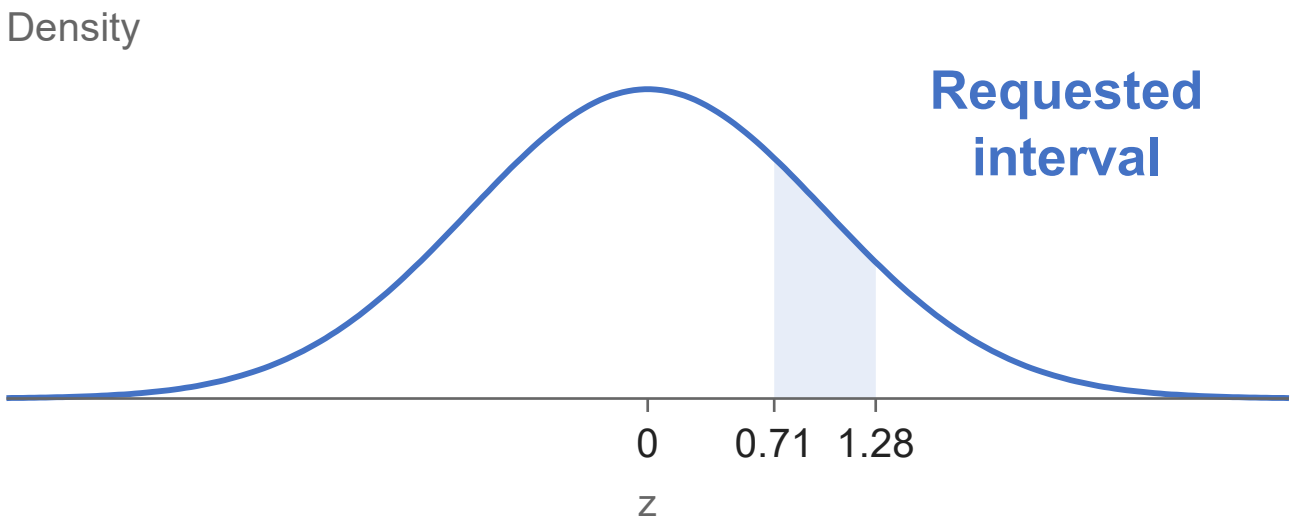
Decide whether 0.75 is a table entry or a row/column label.

A probability is an area; a percentile is the measurement or z-score at that cumulative area.

Example 9(a): subtract cumulative areas

THE AREA BETWEEN TWO BOUNDS IS THE LARGER CDF MINUS THE SMALLER CDF

$$P(0.71 < Z < 1.28) = \Phi(1.28) - \Phi(0.71)$$



Φ(1.28) ≈ 0.8997
Φ(0.71) ≈ 0.7611

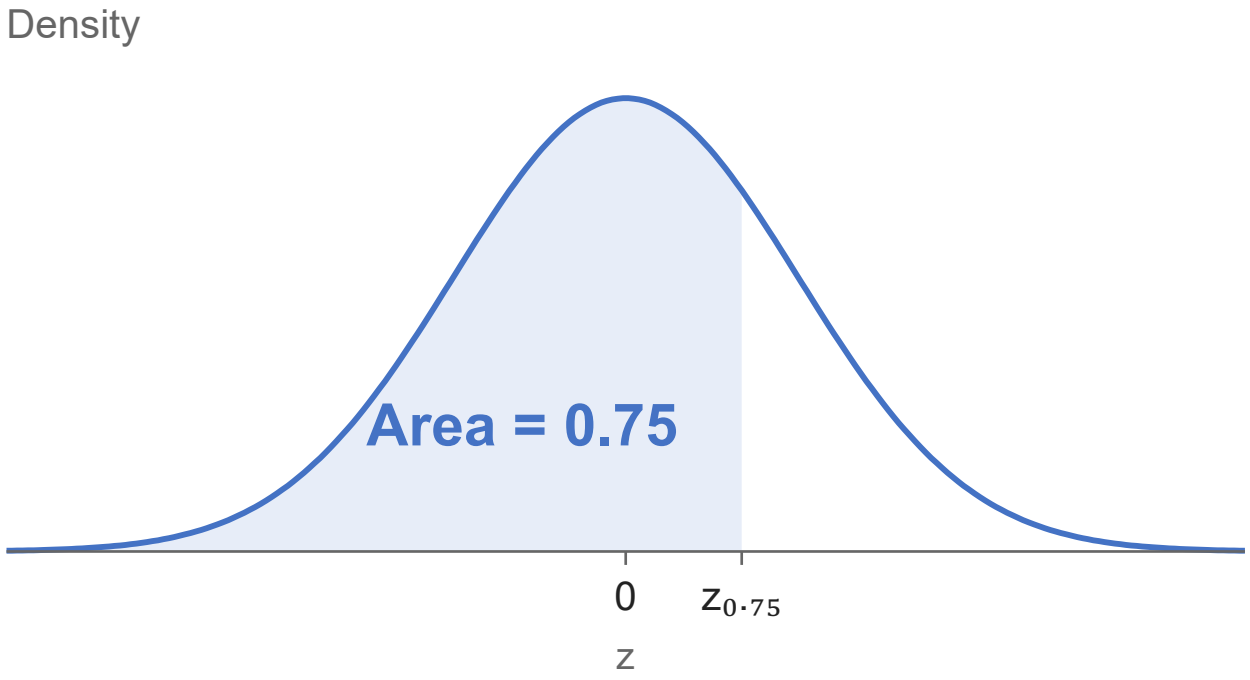
$$\approx 0.8997 - 0.7611 = 0.1386$$

Area ≈ 13.86%.

Do not add the two table entries: both include much of the same left-hand area.

Example 9(b): the 75th percentile is a z-score

LOOK FOR 0.7500 INSIDE THE TABLE, THEN READ THE z LOCATION



$$\Phi(z_{0.75}) = 0.75$$

z	Φ(z)
0.67	0.7486
0.68	0.7517

$$z_{0.75} \approx 0.6745$$

A two-decimal table gives about 0.67; the numerical percentile is about 0.6745.

Return to original units when the question requires it

FORWARD: BOUNDS → AREAS | INVERSE: CUMULATIVE AREA → A BOUND

1. Translate measurement bounds into standard units.

$$P(a < X < b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

2. For a percentile, recover the measurement after finding z.

$$x_p = \mu + \sigma z_p, \qquad \Phi(z_p) = p$$

Here p is a cumulative probability between 0 and 1.

For a negative z, symmetry also gives $\Phi(-z) = 1 - \Phi(z)$.

Estimating the normal-model parameters

USE THE SAMPLE MEAN FOR μ AND THE SAMPLE STANDARD DEVIATION FOR σ

Assume independent observations from the same normal population.

$$\widehat{\mu} = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Sample mean → population centre

$$\widehat{\sigma} = s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

Sample SD → population spread

$$E[\overline{X}] = \mu$$

The sample mean is unbiased for μ.

A sample estimate is not the same as a known population parameter.

Standard deviation and standard error are different

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DISTINGUISH VARIABILITY OF ONE OBSERVATION FROM VARIABILITY OF AN AVERAGE

One measurement

$$\mathrm{SD}(X_i) = \sigma$$

Spread of individual values around the population mean.

Mean of n measurements

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

When σ is unknown, estimate it by:

$$\widehat{\mathrm{SE}}(\overline{X}) = \frac{s}{\sqrt{n}}$$

Use σ for the z-score of one observation; σ/√n describes the spread of a sample mean.

A linear transformation preserves normality

IF $X \sim N(\mu, \sigma^2)$, LET $Y = aX + b$; $a \neq 0$ AND b ARE CONSTANTS

$$Y = aX + b \sim N(a\mu + b,\ a^2\sigma^2)$$

MEAN	$\mu_Y = a\mu + b$	Keep the sign of a; add b.
VARIANCE	$\sigma_Y^2 = a^2\sigma^2$	Square a; the shift b disappears.
STANDARD DEVIATION	$\sigma_Y = \lvert a \rvert \sigma$	Use |a| so spread remains nonnegative.

Adding a constant shifts the distribution; it does not add to the variance or SD.

Example 10: convert a normal temperature measurement

THE QUESTION ASKS FOR A DISTRIBUTION—NOT ONLY A CONVERTED MEAN

A chemist measures the temperature of a solution in °C.
The measurement C is normal with mean 40 °C and SD 1 °C.

$$C \sim N(40, 1^2), \qquad F = 1.8C + 32$$

What is the distribution of F?

State its family, mean, variance, and standard deviation.

Before calculating: which of the quantities changes by adding 32?

Example 10: normal in Fahrenheit, with a new scale

WORKED SOLUTION DERIVED FROM THE SOURCE CONVERSION

MEAN $\mu_F = 1.8(40) + 32 = 104$ °F

VARIANCE $\sigma_F^2 = (1.8)^2(1)^2 = 3.24$ (°F)²

STANDARD DEVIATION $\sigma_F = |1.8|(1) = 1.8$ °F

$$F \sim N(104, 3.24)$$

F ~ N(104, 3.24) uses variance 3.24—not standard deviation 1.8—in its second position.

Linear combinations of independent normal variables

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STATE INDEPENDENCE BEFORE USING THIS DISTRIBUTION RULE

Let X_i be independent normal variables with means μ_i and variances σ_i^2.

Define $W = c_1X_1 + \cdots + c_nX_n$, with at least one nonzero coefficient.

$$W \sim N\left(\sum_{i=1}^{n} c_i\mu_i,\ \sum_{i=1}^{n} c_i^2\sigma_i^2\right)$$

MEAN: RETAIN COEFFICIENT SIGNS

$$\mu_W = \sum_{i=1}^{n} c_i\mu_i$$

VARIANCE: SQUARE COEFFICIENTS

$$\sigma_W^2 = \sum_{i=1}^{n} c_i^2\sigma_i^2$$

Independence is a sufficient condition for this normal law and its displayed variance.

Independent normal sums and differences

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A MINUS SIGN CHANGES THE MEAN—NOT THE SIGN OF AN INDIVIDUAL VARIANCE

Assume X and Y are independent normal random variables.

SUM $$X + Y \sim N(\mu_X + \mu_Y,\ \sigma_X^2 + \sigma_Y^2)$$

DIFFERENCE $$X - Y \sim N(\mu_X - \mu_Y,\ \sigma_X^2 + \sigma_Y^2)$$

For X − Y, the coefficient on Y is −1; its squared variance coefficient is still +1.

The mean of a normal sample is exactly normal

THIS RESULT DOES NOT REQUIRE A LARGE-SAMPLE APPROXIMATION

$$X_1, \ldots, X_n \text{ independent}, \qquad X_i \sim N(\mu, \sigma^2)$$

$$\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Mean stays at μ.
Variance becomes σ²/n.
SD becomes σ/√n.

$$Z_{\overline{X}} = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

The denominator is the SD of the variable being standardized: σ for X, σ/√n for $\overline{X}$.

Assessing normality: evidence, sample size, and caution

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SOURCE CAUTION, WITH AN EXPLICIT MODEL-CHOICE CLARIFICATION

LARGER SAMPLES

A histogram may indicate whether a roughly normal shape is plausible.

Probability plots are another tool (source preview: §4.10).

SMALL SAMPLES

Shape is harder to assess.

Absence of obvious outliers does not establish normality.

Use knowledge of the process.

OUTLIERS

The source warns against uncritical normal modelling when outliers are present.

Investigate the observation, context, and model fit.

Clarification: an outlier alone neither proves non-normality nor establishes a lognormal model.

The lognormal distribution: normal after taking logarithms

THE SOURCE DEFINES A TRANSFORMATION—NOT A RULE TO USE FOR EVERY OUTLIER

NORMAL VARIABLE

$$X \sim N(\mu, \sigma^2)$$

exp →

POSITIVE, LOGNORMAL VARIABLE

$$Y = e^X > 0$$

$$\ln Y = X \sim N(\mu, \sigma^2)$$

Use the natural logarithm, ln, to move back to normal units.

The parameters μ and σ² describe ln(Y), not the mean and variance of Y itself.

Lognormal density: the parameters are on the log scale

USE y > 0; THE DENSITY IS ZERO ELSEWHERE

$$f_Y(y) = \begin{cases} \dfrac{1}{\sigma y \sqrt{2\pi}} \exp\left[-\dfrac{(\ln y - \mu)^2}{2\sigma^2} \right], & y > 0, \\ 0, & y \le 0. \end{cases}$$

$$E[\ln Y] = \mu, \qquad \mathrm{Var}(\ln Y) = \sigma^2$$

The factor 1/y distinguishes this density from a normal density in y.

Do not substitute a raw-scale mean and variance for μ and σ² without converting the model.

Lognormal probabilities: transform back to normal units

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DERIVED APPLICATION OF THE SOURCE RELATION X = ln(Y)

For a positive threshold y, logarithms preserve the order of values.

$$P(Y \le y) = P(\ln Y \le \ln y)$$

$$= \Phi\left(\frac{\ln y - \mu}{\sigma}\right) \quad (y > 0)$$

$$F_Y(y) = 0 \quad \text{for } y \le 0$$

Take ln(y) first; then subtract the log-scale μ and divide by the log-scale σ.

Optional reference: lognormal mean and variance

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DERIVED SUPPLEMENT — ORIGINAL SLIDE 20 DISPLAYS THE PDF ONLY

$$E[Y^r] = E[e^{rX}] = \exp\left(r\mu + \frac{r^2\sigma^2}{2}\right)$$

Set r = 1 and r = 2:

$$E[Y] = e^{\mu+\sigma^2/2}$$

$$E[Y^2] = e^{2\mu+2\sigma^2}$$

$$\mathrm{Var}(Y) = (e^{\sigma^2} - 1)e^{2\mu+\sigma^2}$$

Raw-scale moments are not μ and σ². The derivation is in the speaker notes.

Normal and lognormal: keep the variable and scale straight

THE RELATIONSHIP IS EXPONENTIAL, NOT A LINEAR CHANGE OF UNITS

Feature	Normal X	Lognormal Y
Possible values	All real values	Positive values only
Normal quantity	X itself	ln(Y)
Parameter μ	Mean of X	Mean of ln(Y)
Parameter σ^2	Variance of X	Variance of ln(Y)
Probability route	Standardize x	Take ln(y), then standardize

Before interpreting a parameter, ask: “Is this a raw measurement or a logarithm?”

Common mistakes: repair the interpretation first

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CHECK THE PARAMETER, THE TAIL, THE TRANSFORMATION, AND THE VARIABLE

01 **Variance mistaken for SD**

$N(\mu, \sigma^2)$ stores variance; divide by σ when standardizing.

02 **Every table entry treated as the answer**

A left-tail entry needs a complement for a right-tail request.

03 **A percentile confused with a probability**

The 75th percentile is a z or x location, not the number 0.75.

04 **Spread shifted or variances subtracted**

A shift does not change spread; square every coefficient.

05 **The wrong scale used for lognormal parameters**

μ and σ^2 describe $\ln(Y)$, not Y.

A correct calculation starts with a correct description of what the number represents.

Concept check: explain why each statement is wrong

DISCUSS BEFORE REVEALING THE NEXT SLIDE

A Standardizing any random variable makes it normally distributed.

B The area to the right of $z = 1.38$ is $\Phi(1.38)$.

C For independent normal X and Y, $\mathrm{Var}(X - Y) = \sigma_X^2 - \sigma_Y^2$.

D A lognormal variable with parameters μ and σ^2 has mean μ.

Replace each claim with a correct statement; a one-word “false” is not an explanation.

Concept check: corrected statements

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EACH REPAIR RESTORES A CONDITION OR A MEANING

A **If X is normal, its standardized variable Z is standard normal.**

Subtracting and scaling do not, by themselves, establish a normal model.

B **$P(Z > 1.38) = 1 - \Phi(1.38) \approx 0.0838$.**

The table is left-tail; the requested event is right-tail.

C **$\mathrm{Var}(X - Y) = \sigma_X^2 + \sigma_Y^2$ under independence.**

The coefficient of Y is -1, and $(-1)^2 = 1$.

D **For the lognormal parameterization, $E[\ln(Y)] = \mu$.**

The parameter is on the logarithmic scale, not the raw scale.

State the model and assumptions before substituting numbers into a formula.

Lecture 11: key takeaways and questions

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NORMAL MODEL → STANDARD UNITS → AREA OR PERCENTILE → INTERPRETATION

1. **Read N(μ, σ²) carefully.**
   The second parameter is variance; standardization uses SD.
2. **Sketch the probability event.**
   Left tail, right tail, or interval determines the table operation.
3. **Reverse the workflow for percentiles.**
   Find the cumulative area's z-score, then restore original units.
4. **Keep assumptions visible.**
   Independence supports the displayed normal sum and mean rules.
5. **Keep lognormal parameters on the log scale.**
   ln(Y), not Y, is normal with mean μ and variance σ².

> Questions? Be ready to explain the variable, the parameters, the area, and the assumptions.